Tayyab Ur Rehman

Full Stack Developer | AI Platforms, Next.js, FastAPI & AWS

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Summary

Full stack developer with hands-on experience taking AI-powered web platforms from raw idea to high-scale production, framing the problem, designing the system, owning the architecture, and shipping and serving to real users. I build end to end: React, Next.js, Remix and TypeScript on the frontend; FastAPI and PostgreSQL on the backend; deployed on AWS with Docker and CI/CD. Lead engineer on Mishkat VLE (a multi-branch school LMS) and a core builder on Xyova (an AI evaluation platform). My entire development workflow is fully agentic and automated: spec-first design, AI coding agents (Kiro, Claude Code, Cursor) orchestrated over the Model Context Protocol, automated tests and CI/CD, giving me a **10x output multiplier** while I stay focused on architecture, review, and delivery.

Leadership & How I Build

- **Idea to production:** take features from a rough spec to a served, monitored system: problem framing, data model, build, deployment, and iteration with real users across multiple production platforms.
- **Technical leadership:** lead engineer on Mishkat VLE: own the architecture and data model, define frontend/API contracts, review the team's work, and drive delivery under ambiguity.
- **System design & architecture:** own technical calls on multi-tenant platforms: auth / RBAC, layered backends (endpoint-service-repository), real-time media, and build-vs-buy (self-hosted LiveKit for live video at school scale).
- **Agentic, automated workflow, 10x output:** spec-first design with AI coding agents (Kiro, Claude Code, Cursor) orchestrated over MCP, automated testing, and CI/CD; agents run the implementation loop while I own architecture, review, and quality.

Core Skills

- **Languages:** JavaScript, TypeScript, Python, HTML5, CSS3.
- **Frontend:** React.js, Next.js, Remix.js, Redux, React Query, Chart.js, LiveKit, WebSockets.
- **Backend & APIs:** FastAPI, Node.js, REST APIs.
- **Cloud & DevOps:** AWS (EC2, ECS, Lambda, S3, SQS, SNS, Cognito, SES, CloudWatch), Docker, GitHub Actions, CI/CD, Vercel.
- **Databases:** PostgreSQL, Supabase, Prisma ORM, PL/pgSQL.
- **Styling & CMS:** Tailwind CSS, Bootstrap, Shopify Polaris, Custom CSS, WordPress.
- **Architecture:** system design, multi-tenancy / RBAC, API design, technical leadership.
- **AI-native Delivery:** agentic coding (Kiro, Claude Code, Cursor), MCP servers and tooling, spec-driven development, automated testing & CI/CD.

Experience

Full Stack Developer

Oct 2025 – Present

Mishkat Quantum Labs | Karachi, Pakistan (Onsite)

- Led architecture, data models, frontend/API contracts, and key product decisions across Mishkat VLE, Atlas (SIS) and Xyova, taking each from idea to production.
- **Live classes:** delivered real-time virtual classrooms with self-hosted LiveKit and WebSockets, including presence-driven room join, participant controls, whiteboard sync and persistence, and mobile responsiveness in the Next.js / TypeScript app.
- **Learning features:** built homework and classwork (text and attachments), assessments with school-wide mark categories and multiple grading modes, timetable file management, attendance, and multi-section student membership, connecting Next.js views to FastAPI services end to end.
- **Platform & data:** designed a layered FastAPI and PostgreSQL service architecture, with AWS S3 for shared file lockers and attachments and Cognito for auth; shipped resubmit-after-scoring, server-side pagination, recycle bin with cascade delete, and user disable-access with delete-conflict handling.
- **Atlas (School Information System):** migrated a legacy SIS prototype to Next.js, TypeScript and Supabase and built admissions, enquiries, billing, finance / collections / treasury, fee-counter lookup, audit and notifications; deployed with nginx and PM2.

- **Xyova (Next.js + Tailwind):** built z-score analytics charts and dashboards with Chart.js; shipped rubric activate/deactivate, a “needs human review” flag, org-admin minutes & subscription flows, OTP/auth flows, and a full landing-page redesign; managed the v1.5 release.
- **AI Co-Teacher (lesson planning):** delivered lesson-plan generation and audio-player integration, with class, section and teacher assignment.
- Automated deployment pipelines and CI/CD with Docker, GitHub Actions, and AWS (EC2, S3, Lambda, ECS); worked with AI backend engineers on MCP servers and tuned Turbopack builds to keep delivery reliable.

Frontend Developer
May 2025 – Apr 2026

Crestotech | Islamabad, Pakistan (Remote)

- Built and maintained a Shopify B2B wholesale pricing platform with Remix, React.js, Shopify Polaris, and Prisma, covering custom wholesale price lists, tiered pricing, and merchant admin flows.
- Delivered client-facing pages, controllers, and dashboard flows in a SaaS React codebase and a Laravel-backed app.
- Converted Figma handoffs into responsive, high-quality interfaces with custom CSS, and delivered WordPress theme-based sites for clients.
- Engineered scalable, reusable UI components with React.js and Tailwind CSS, focused on performance and accessibility.

Frontend Developer
Nov 2024 – Apr 2025

NestedFor Software | Islamabad, Pakistan

- Built responsive production UIs in React.js for immigration-clinic and travel-booking platforms.
- Developed a job portal end to end on the MERN stack: responsive admin dashboard (sidebar, cards, data tables), job list / detail / view pages, applicant and shortlisting flows, user and provider management, and profile and auth sections.
- Converted design specifications into maintainable code and reusable components for seamless cross-device experiences.

Frontend Developer (Internship)
Feb 2024 – Jun 2024

CodeSoft

- Built real-world frontend logic and reusable, responsive, performance-optimized components with JavaScript, React.js, and Redux.

Selected Projects

- **Mishkat VLE** (mishkatvle.com): lead engineer on a multi-branch, AI-powered school LMS: real-time live classes over self-hosted LiveKit, homework and assessments, timetable, shared lockers, and role-based portals. Next.js + FastAPI + PostgreSQL on AWS.
- **Xyova** (xyova.com): an AI evaluation platform turning recorded lessons into audited, scored insights with z-score analytics, rubrics, and a human-review workflow. Next.js + Tailwind frontend on AWS.
- **Support System** (support-system-mql.vercel.app): a multi-project support ticket platform built end to end. Full ticket lifecycle, actor-aware email notifications (Jira-style), comments with @mentions, S3 file attachments, full-text search, analytics dashboards, role-based access, and a complete audit trail. FastAPI + Next.js + AWS (Cognito, S3, SES, CloudWatch), Dockerized with optional Jira & Discord.
- **MCP Servers (GitHub, Jira & Discord)**: Model Context Protocol servers that let AI clients work with repositories, manage projects and issues, and communicate over a standardized protocol.

Education, Certifications & Writing

Bachelor of Computer Science, University of Agriculture, Peshawar (2020 – 2024)

AWS Certified Cloud Practitioner (Associate)

Writing: Building MCP Servers (GitHub, Jira & Discord) | Integrating LiveKit WebSockets with Next.js for Real-Time Communication